

RUTWIK PATEL

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EDUCATION

University of Southern California, Los Angeles, CA

August 2023 - May 2025

Masters of Science in Computer Science

GPA: 3.81/4.0

University of Mumbai, Mumbai, India

August 2019 - May 2023

Bachelor of Technology in Information Technology

GPA: 3.8/4.0

EXPERIENCE

Software Engineer

Sept 2025*

Sigma Computing

New York City, NY

- Delivered formula-based conditional background styling for workbook containers, enabling dynamic visual insights.
- Implemented Form_V2 enhancements (delete fields, read-only mode) in TypeScript, reducing UI friction for end-users.
- Added automated test (unit + Cypress) for new UI flows, preventing regression issues across edit/view modes.

Software Engineer Intern

July 2024 - August 2025

World Salon

Los Angeles, CA

- Spearheaded Event creation workflow using MERN stack with JWT authentication, role-based access control
- Architected microservices for payment/user modules with React Router, Redux improving performance by 20%
- Worked with AI team to build Puppeteer scraper extracting 39000+ speaker profiles and automated email workflows
- Leveraged prompt engineering and OpenAI APIs to automate speaker profiling and personalized outreach emails
- Deployed containerized app on AWS S3/EC2 with GitHub Actions CI/CD pipeline, Jest testing, and CloudWatch

Research Assistant

February 2024 - May 2025

USC Marshall School of Business

Los Angeles, CA

- Built automated web scraper using Selenium to collect 10 years of SP1500 sustainability reports (15K+ PDFs)
- Developed RAG system with 10K+ vector embeddings and semantic search across 500GB+ energy transition data
- Executed multi-threaded PDF processing pipeline with OCR support attaining 30% faster extraction rates
- Created PostgreSQL database indexing 2M+ data points with React UI for researchers accessing processed reports
- Automated Orbis database scraper for 1.8M+ companies using dynamic threading, reducing manual workload by 90%

Software Development Engineer Intern

May 2022 - July 2022

Smart Consultant

Mumbai, India

- Developed inventory management system for small retail businesses with Django REST API and PostgreSQL database
- Delivered dashboard analytics showing sales trends, profit margins, and automated ABC analysis for 5000+ SKUs
- Deployed on AWS EC2 achieving 200ms response times with S3/CloudFront CDN, supporting 15+ concurrent clients

ACADEMIC PROJECTS

RAG QA App ([Github](#)) | Engineered RAG-powered QA tool using Streamlit and LangChain with Chroma vector database for semantic document search. Processed PDF/DOCX/TXT files into 512-token chunks enabling sub-second retrieval across private document corpora. Optimized LangChain summarization with parallel map-reduce chunking, reducing embedding API calls by 30% and improving query latency by 20% through batch processing strategies.

Leveraging BERT and RoBERTa for Fake News Detection | Utilized deep learning models (CNN-LSTM, BERT, and RoBERTa) to detect fake news, with RoBERTa achieving over 93% accuracy. Evaluated models on WELFake and Kaggle datasets (115K+ articles), demonstrating RoBERTa's superior precision, recall, and F1-score (all ~94%)

Stock Insight Application ([Live Demo](#)) | Designed full-stack stock trading platform on GCP with MEAN stack with Finnhub and Polygon APIs for real-time market data. Implemented interactive Highcharts with technical indicators. Achieved 96/100 performance score through lazy loading, responsive Angular Material design, and optimized API caching.

Cataract Detection with Explainable AI (XAI) ([Github](#)) | Led team of 3 to develop CNN-based cataract detection system achieving 97% accuracy with explainable AI integration. Integrated GradCAM for visualizing model decisions and localizing affected regions in eye images through heatmap generation. Built Streamlit web interface for medical professionals to upload eye scans and receive interpretable predictions with confidence scores and visual explanations.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Swift | **Databases:** PostgreSQL, MongoDB, MySQL, Chroma, Pinecone

Web Development: Node, React.js, Angular.js, Express.js, Django, Flask, CI/CD, Terraform

ML Libraries: PyTorch, TensorFlow, LangChain, Keras | **DevOps:** AWS(S3, Cloudfront, Route, EC2), Docker

PUBLICATIONS

- Published 2 IEEE papers: federated learning ([SICTIM 2023](#)) and cataract detection ([PuneCon 2023](#)).